

Marcelo Morales Gómez

Columbus, Ohio

787-579-8044 | morales.374@osu.edu

Education

The Ohio State University

Ph. D. in Computer Science and Engineering

- Member of the World 3 Cybersecurity Innovations Lab (W3CIL)

Columbus, Ohio

August 2022 - Current

Ohio University

B.S. in Computer Science

- Graduated with Magna Cum Laude
- Member of the Honors Tutorial College
- Independent Study:** Blockchain & Cryptocurrency Technologies, Digital Signal Processing, Network on Chip, Computer Security, Advanced Problem Solving

Athens, Ohio

May 2018 - May 2022

Work Experience

The Ohio State University

Graduate Research Assistant

- Researched blockchain security with a focus on smart contracts.
- Developed a prototype using agent-based simulation to detect smart contract vulnerabilities by modeling economic environments.
- Applied machine learning to analyze and enhance the security of blockchain oracles.
- Collaborated with other research labs on blockchain security projects.

Columbus, Ohio

May 2023 - Current

The Ohio State University

Graduate Teaching Assistant

- Graded labs and exams for a C++ programming course.
- Held office hours to assist students with course material and debugging.
- Clarified complex programming concepts to improve student understanding.

Columbus, Ohio

August 2022 - May 2023

JPMorgan Chase & Co.

Software Engineer Intern

- Developed a Java Spring Boot API interacting with DynamoDB on an EKS cluster.
- Utilized Jenkins for CI/CD and Terraform for AWS infrastructure provisioning.
- Built a React front-end for API interaction and optimized performance using JMeter.
- Achieved 100% code coverage with JUnit tests.
- Documented API functionality in Confluence and contributed to daily scrum meetings.
- Technical Skills:** Java, AWS, React, Jenkins, Terraform.

Columbus, Ohio

June 2022 - August 2022

JPMorgan Chase & Co.

Software Engineer Intern

- Developed APIs and microservices using Java Spring Boot.
- Tested REST services with Postman and analyzed logs via Splunk.
- Wrote unit tests (JUnit) and integration tests (Mockito).
- Created SQL scripts for interactions with Oracle SQL database.
- Collaborated with vendor team for product backlog refinement.
- Participated in Sprint reviews and presented product demos.
- Led Scrum meetings and managed agile boards using Jira.
- Technical Skills:** Java, Python, AWS.

Columbus, Ohio

June 2021 - August 2021

Publications

Your Exploit is Mine: Instantly Synthesizing Counterattack Smart Contract

Zhuo Zhang, Zhiqiang Lin, **Marcelo Morales**, Xiangyu Zhang, Kaiyuan Zhang

32nd USENIX Security Symposium (USENIX Security 23), 2023, Anaheim, CA

University Projects

Proofmatic Project

Athens, Ohio

Ohio University

August 2021 - May 2022

- Served as Scrum Master, team lead, and full-stack developer for the Proofmatic project, overseeing team coordination and project timelines.
- Developed an advanced web-based platform to support the learning of mathematical proofs, featuring integrated development environments (IDEs) for LaTeX and Coq.
- Participated in client meetings to gather requirements and provide project updates, ensuring alignment with client expectations throughout the development cycle.
- **Technical Skills Utilized:** Node.js, TypeScript, React, LaTeX, Coq

CatMap

Athens, Ohio

Ohio University

August 2019 - December 2019

- Developed an indoor navigation app designed to help students locate and navigate to classrooms within complex campus buildings.
- Integrated building maps and room data to provide turn-by-turn real-time directions for easier navigation.
- Collaborated with a team to design user-friendly interfaces and ensure accurate mapping of campus facilities.
- **Technical Skills:** Java, Mobile App Development

Skills

Programming Python, C++, JavaScript, Typescript, React, Solidity

Miscellaneous Linux, Shell (Bash/Zsh), $\LaTeX$ (Overleaf/R Markdown) , Microsoft Office, Git.

Soft Skills Time Management, Teamwork, Problem-solving, Engaging Presentation.

Achievements

2020 **2nd Place**, (Kubernetes Server Manager) Revolution UC

University of

Cincinnati

2019 **Best Environmental Hack**, Hackaplachia

Ohio University

Languages

English Bilingual Proficiency

Spanish Native Proficiency

References available upon request.